CMPG 311
Group 12 Project PHASE 1
18 March 2025
Topic: Clinic Appointment System

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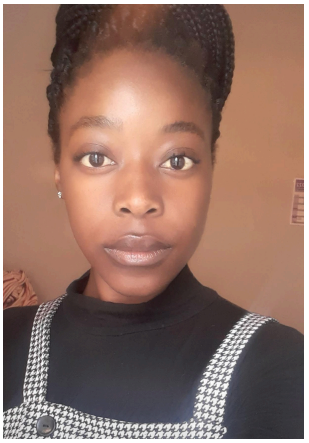
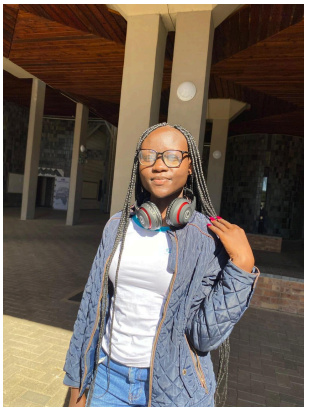
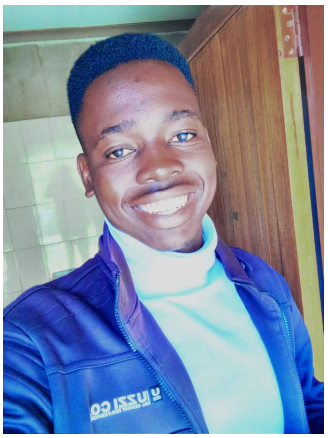
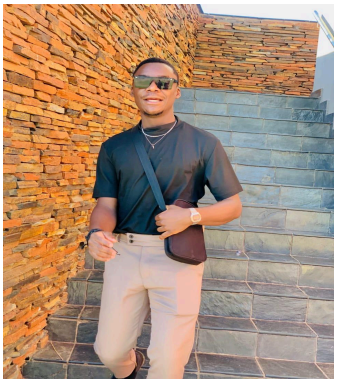
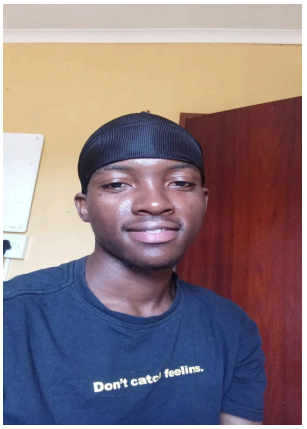
Ms. Masinga MS	44527802		Cell: +27 73 913 6169 Email: 44527802@mynwu.ac.za
Ms. Munyai MG	42551900		Cell: +27 72 110 5380 Email: 42551900@mynwu.ac.za
Mr. Chauke DX	42705614		Cell: +27 79 867 8786 Email: 42705614@mynwu.ac.za

Mr. Khoza S	32462409		Cell: +27 76 602 6847 Email: 32462409@mynwu.ac.za
Mr. Mabasa NV	47337354		Cell: +27 67 636 8768 Email: ntshembho17@gmail.com

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- **MISSION**

The mission of the clinic is to provide high-quality healthcare services to patients in an efficient and organized manner. The clinic aims to streamline its operations by implementing a database system that automates appointments, manages patient records, and improves overall service delivery. The appointment system aims to provide a way

of scheduling, managing, and tracking appointments for both patients and healthcare professionals. This system will allow patients to book appointments based on the doctor's availability, also help health professionals manage their schedules, and enable administrative staff to track appointments and patient details.

OBJECTIVES :

- To allow patients to book appointments easily through an online interface.
- To enable doctors to manage their availability and view their schedules.
- To provide an efficient way to manage patient information and appointment histories.
- To ensure smooth and secure data handling, using PostgreSQL for patient data storage.
- Develop a centralized appointment scheduling system that eliminates double bookings and scheduling conflicts
- Implement automated resource allocation to optimize the utilization of rooms and equipment
- Create a unified patient data repository that integrates with existing systems
- Enable data-driven decision-making through comprehensive reporting capabilities
- Reduce administrative burden by automating routine scheduling tasks
- Decrease average patient wait times by 25% within six months of implementation

Strategic Goals:

- **Patient Care Excellence** which delivers high-quality medical services with minimal waiting times and maximum patient satisfaction.
- **Operational Efficiency**: Streamline administrative and clinical workflows to optimize resource utilization.
- **Financial Sustainability**: maximizing profit but keeping healthcare services affordable and reliable
- **Technological Advancement**: Implement electronic health records (EHR) and telemedicine to enhance patient management.
- **Community Outreach**: Conduct awareness programs, free check-ups, and health education seminars.
- Staff Development: To foster a supportive work environment that encourages professional growth and job satisfaction among employees.

Description of Operations

The clinic operates as a multi-specialty healthcare facility, by offering various medical services, diagnostics, and outpatient care.

- Patient Scheduling: Patients can book, reschedule, or cancel appointments online.
- Doctor Availability: Doctors can set their availability, and the system automatically allocates appointments based on their schedules.
- Administrative Tasks: Staff can manage patient records, doctor schedules, and generate reports.
- Notifications: Automated reminders (email/SMS) for upcoming appointments.

- Billing Integration: Optional integration with billing systems for payment processing.
- Scheduling Appointments
- Online appointment scheduling and time slot viewing are offered to patients.
- Physicians are able to schedule appointments and control their availability.
- Administrative personnel can handle doctor's schedules or make appointments on behalf of patients.
- Automated Notifications
- Patients are automatically reminded of upcoming appointments via SMS or email.
- The ability to use notifications to confirm, reschedule, or cancel appointments.
- Management of Patient Information
- Secure storage of patient medical records and personal information.
- the capacity to keep track of each patient's appointment history.
- Electronic Health Record (EHR) system integration is an optional feature.
- Management of Waitlists
- Patients may be placed on a waiting list if there are no open spots.
- Notify patients automatically when slots become available.
- Secure Login/Authentication
- Role-based authentication for users: Admin, Doctors, and Patients.
- User Dashboard

- Personalized dashboards for both patients and doctors that shows upcoming appointments, medical records, and available time slots.

ORGANIZATIONAL STRUCTURE

The clinic follows a hierarchical structure to ensure smooth operations and efficient service delivery.

Leadership & Administration

a. Clinic Director (CEO): Oversees overall strategy, financial management, and clinic operations.
b. Medical Director: Responsible for clinical governance, compliance, and patient safety.
c. Operations Manager: Manages daily activities, staff scheduling, and logistics.
d. Finance & Accounts Team: Handles billing, insurance claims, and financial planning.
e. HR & Administrative Team: Manages hiring, training, and staff welfare. Medical Staff
f. General Practitioners (GPs): Provide primary healthcare services.
g. Specialists: Cardiologists, pediatricians, gynecologists, etc. • Nurses: Assist in patient care, administer medications, and handle emergency cases.
h. Laboratory Technicians: Conduct medical tests and diagnostics.
i. Pharmacists: Dispense medications.
j. Support Staff • Receptionists & Front Desk Officers: Manage appointments, inquiries, and patient records.
k. IT Support Team: Maintains digital systems, online bookings, and telemedicine services.

I. Cleaning & Maintenance Staff: Ensures hygiene and upkeep of the facility.

PROBLEMS AND CONSTRAINTS

Identified Problems

- Manual scheduling inefficiencies leading to appointment overlaps.
- Patients and doctors/nurses might accidentally double book or miss appointments leading to Scheduling conflicts.
- Limited access to real-time doctor availability, causing patient dissatisfaction.
- Lack of patient history tracking, making follow-ups difficult.
- Paper-based records are prone to loss and errors.Technological Gaps: Outdated systems and limited telemedicine capabilities.
- Community Engagement: Low awareness and participation in outreach programs.

Constraints

- The system must comply with POPIA (Protection of Personal Information Act) in South Africa for data privacy.
- Doctors' schedules may change, requiring real-time database updates.
- System accessibility should be user-friendly, considering patients with limited technical skills.
- **Budget Limitations**: The clinic operates with restricted financial resources for technology implementation.
- **Legacy Systems Integration**: Any new database system must integrate with existing billing and electronic health record systems.

- **Staff Technical Proficiency**: Varying levels of technical expertise among staff will require user-friendly interfaces and comprehensive training.
- **Regulatory Compliance**: The system must adhere to healthcare regulations including Protection of Personal Information Act (POPIA) for patient privacy and data security.
- **Minimal Disruption Requirement**: Implementation must occur with minimal impact to ongoing patient care operations.
- **Scalability**: As the clinic grows, the system might not be able to handle more users being added.

DATABASE SYSTEM SPECIFICATIONS

System Objectives

The database system is designed to address the constraints and problems identified in the clinic's operations. Its primary objectives are:

- Reduce Waiting Times – Efficient appointment scheduling and patient record retrieval.
- Improve Patient Data Management – Secure storage and quick access to medical records.
- Enhance Inventory Management – Track availability of medications and medical supplies.
- Streamline Billing & Payments – Automate invoice generation and track pending payments.
- Ensure Compliance with Regulations – Secure data storage, access control, and audit logs.
- Enable Remote Access – Support telemedicine and online consultation records.
- Provide real-time tracking of doctor availability.

- Centralize patient records and appointment history for easy access.
- Implement automated reminders to reduce missed appointments.
- Optimize doctor schedules to prevent overbooking or underutilization.

Information required from the database

1. **Patient information:**

   Patient ID (Primary Key)

   Full Name

   Date of Birth

   Contact Information

   Address

   Medical History

   Allergies

   Insurance Details

2. **Appiontment records**

   Appointment ID (Primary Key)

   Patient ID (Foreign Key)

   Doctor ID (Foreign Key)

   Date & Time

   Status (Scheduled, Completed, Canceled)

   Consultation Notes

3. **Medical Staff Information**

   Doctor/Nurse ID (Primary Key)

   Name

   Specialization

   Contact Information

   Work Schedule

4. **Medical Records**

   Record ID (Primary Key)

   Patient ID (Foreign Key)

   Doctor ID (Foreign Key)

   Diagnosis

   Prescriptions

   Lab Test Results

   Treatment Plan

5. **Pharmacy & Inventory Management**

   Item ID (Primary Key)

   Name of Medicine/Supply

   Stock Quantity

   Expiry Date

   Supplier Information

   Reorder Level

6. **Billing & Payments**

   Invoice ID (Primary Key)

   Patient ID (Foreign Key)

   Date of Billing

Total Amount

Payment Status (Paid, Pending, Overdue)

Payment Method (Cash, Card, Insurance)

SCOPE

The database will cover the following functional areas within the clinic:

Included in the Scope:

- Patient Registration & Record Management – Maintains detailed patient records.
- Appointment Scheduling & Tracking – Manages bookings with doctors.
- Medical Consultation & Treatment Records – Stores diagnoses, prescriptions, and lab test results.
- Billing, Payments, and Insurance Processing – Automates financial transactions.
- Pharmacy & Inventory Management – Tracks stock levels to avoid shortages.
- Staff Scheduling & Work Assignments – Manages doctor and nurse shifts.
- Report Generation – Generates patient history, billing reports, and stock reports.

Excluded from the Scope

- Emergency Care System (handled by hospitals, not the clinic).
- In-depth HR Payroll Management (basic staff records included, but payroll managed separately).

- Advanced AI-based Medical Diagnosis (only manual doctor diagnosis stored).

CLEARLY DEFINED BOUNDARIES

The database will have certain limitations to ensure efficiency and compliance.

1. **System Access Restrictions**
    - Doctors & Nurses can access patient records and add consultation details.
    - Receptionists can manage appointments and view billing details. Pharmacy Staff can manage inventory and dispense medication.
    - Administrators have full access, including financial and report generation.
    - Patients can only access their medical records via a secure portal.
2. **Data Security & Compliance**
    - Role-Based Access Control (RBAC) – Users can only access data relevant to their role.
    - Data Encryption – Secure patient records and financial transactions.
    - Backup & Recovery – Regular database backups to prevent data loss. Audit Logs – Track all access and changes made to critical data.
3. **Performance & Scalability**
    - The database handles Up to 10,000 Patient Records Initially (Expandable as the clinic grows).

- Optimized for Fast Queries – Ensures quick retrieval of medical records.

4. **Geographical boundaries**
    - The system will be designed to be used by a single/small network of clinics

GROUP MEMBER CONTRIBUTION

Student number	Phase 1 contribution (%)
40676099	100
39803384	100
38880199	100
44527802	100
42551900	100
42705614	100
32462409	100
47337354	100